

FINAL SUBMISSION

Title of Project: Sorting Machine using Vibration Mechanism

Group No: 2

Team Leader :-Aditya Sambhaji Gaikwad,2022A4PS1001G

ID NO	NAME	CONTRIBUTION
2022A4PS1001G	Aditya Sambhaji Gaikwad	Design & Calculations
2022A4PS1336G	Harsh Lohia	Design & Calculations
2022A4PS1195G	Rishabh Surana	Design & Calculations
2022A4PS1070G	Kunal Warat	CAD Model & Design
2022A4PS0715G	Shashank Prasad	CAD Model & Design
2022A4PS0148G	Pratik Ghosh	Calculations,Report and Design
2022A4PS1092G	Sai Swarup Mishra	Calculations, Report and Design
2022A4PS0930G	Arjit Kumar Singh	Spring Design & Calculations

PROBLEM STATEMENT:

Sorting bolts of varying sizes in a workshop is a time-consuming and labor-intensive task, leading to inefficiencies and errors. Manual sorting often results in mismatches, slows down the assembly process, and increases the risk of damage or loss. As the demand for precise and quick sorting grows, there is a need for a system that can accurately and efficiently categorize bolts according to size. This will improve workflow, reduce human error, and optimize productivity in workshop environments, ensuring that the right components are readily available when needed.

OBJECTIVES:-

- **Accurate Size Classification:** Precisely sort bolts into predefined size categories
- **Material Versatility:** Capable of sorting bolts made from various materials (e.g., steel, stainless steel, brass, aluminum) without the need for recalibration.
- **Minimal Damage:** Handle bolts gently throughout the sorting process, ensuring no bolts sustain any damage or deformation.

USE OF THE MACHINE:-

1. Agriculture and Food Processing:

- Fruit and vegetable sorter

2. Manufacturing:-

- Component sorting

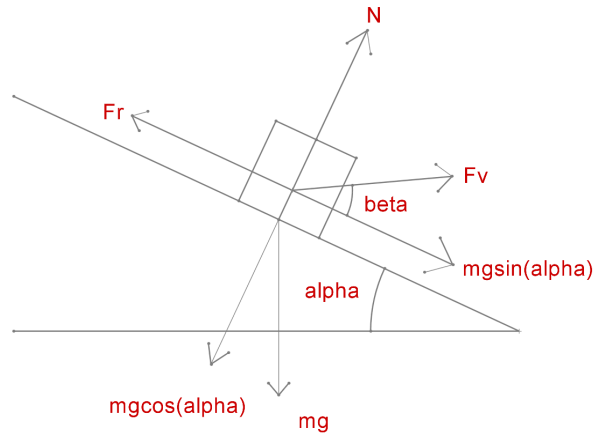
MECHANISM IN PRACTICE :-

The mechanism for the bolt sorting project utilizes a decline ramp with a gradually widening gap to sort bolts by size. Bolts are placed at the top of the ramp, where a vibration tool gently shakes the ramp, causing the bolts to move downward. As the bolts descend, they encounter the widening gap. Smaller bolts fall through the narrower sections first, while larger bolts continue further down the ramp until they reach a section where the gap matches their diameter. Each section of the ramp corresponds to a different bucket, where bolts of the same size are collected, ensuring efficient and accurate sorting.

DESIGN CALCULATIONS:-

REFERENCE	CALCULATIONS	CONCLUSIONS
	<u>Collecting Box Dimensions:</u> The boxes are designed in such a way that only a particular type of bolt falls into each box. There are 5 boxes that are 15 cm each in length. The first box collects every bolt that is smaller than M16 and each box after that collects M16, M18, M20, M22 respectively, and the last box collects all bolts greater than size M22. The plates are angled such that the distance between them varies linearly from 21mm to 36mm.	

Motor Calculation:-



Assuming average weight of 1 bolt=0.09kg

Therefore, weight of 25 bolts =2.25 kg=m

Mass of plate = 5.21kg

Angle of incline of plate with horizontal (α) = 5°

Normal force(N) = $mg \cos(\alpha) = 21.98 \text{ N}$

Minimum μ (so that bolt does not start slipping without vibrating force) = $\tan \alpha$

Minimum $\mu = 0.08$

Static coefficient of friction (μ) = 0.74

$0.74 > 0.08$

Hence, bolt will not slip without vibration

(From Free Body Diagram)

Sinusoidal Motion $x(t) = A \sin (2\pi ft)$ $a(t) = -(2\pi f)^2 A \sin (2\pi ft)$ max a when $\sin (2\pi ft) = 1$	Friction Force: $\mu N = \mu mg \cos \alpha = 16.27 \text{ N}$ Minimum angle of motion with vibrating force required for bolt to lose contact with plate & move upward $= \sin^{-1}(\text{Normal force} / \text{Vibrating force})$ Vibrating Force: $F_v = m \cdot a(v) \text{---(1)}$ where m = mass of bolts Maximum acceleration due to vibration: $a(v) = (2\pi f)^2 \cdot A \text{---(2)}$ where: f = frequency of motion A = Amplitude For a given small vibrating motor f=50 Hz A=0.0005 m $a(v) = (2\pi 50)^2 \cdot 0.0005 = 49.34 \text{ m/s}^2$ $F_v = m \cdot a = 111.03 \text{ N}$ Angle of F_v from the inclined plane = β Min β required for the bolt to lose contact with the plate, $\beta(\text{min}) = \sin^{-1}(21.93/111.03) = 11.49^\circ$ $\therefore$ We assume angle $\beta = 30^\circ$	(approximate values)
--	--	-----------------------------

	From the free body dia, the condition for bolts to move down the plate is $mg\sin(\alpha) + ma(v)\cos(\beta) \geq \mu mg \cos(\alpha)$ Total mass to be vibrated =mass of plate + bolts + motor= $m_i=5.21+2.25/2+0.5=6.835\text{kg}$ Total vibrational force required = $m_i \cdot a(v) = 337.29\text{N}$ Velocity of the plate = Amplitude x Frequency = 0.025m/s Power of motor = $F \cdot v = 8.43\text{W}$ Multiplied with factor of safety $n = 1.25$ so, $P = 10.5 \text{ Watts}$ Link to motor chosen:- <u>3800RPM Electric Motor, Vibration Motor DC 12V Vibrating Motor 10W Carbon Brush Motor w/Adjustable Power Supply for Electric Massage Chair Bed Industry Sieve Shaker More Vibrators - Amazon.com</u> Mass of this motor is 0.5kg It has variable speed control so the frequency can be controlled according to the weight on the plate.	
--	--	--

Reaction forces by columns on plate:

Mass of 1 plate= $m_{\square} = 5.21 \text{ kg}$

Mass of bolts acting on 1 plate= $m_b/2 = 2.25/2 = 1.125 \text{ kg}$

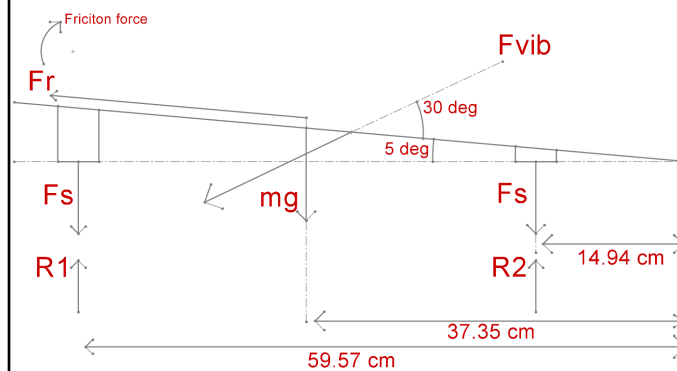
Mass of motor $m_{\square} = 0.5 \text{ kg}$

$m = m_{\square} + m_{\square} + m_b/2 = 6.835 \text{ kg}$

$F_s = 30.2148 \text{ N}$

$F_r = 16.27 \text{ N}$

R_1 and R_2 are loads on each beam



$F_{vib} = 337.29 \text{ N}$

Force Balance Equation:

$$F_{vib} \times \cos 65^\circ + 2(F_s) + mg - F_r \sin 5^\circ = R_1 + R_2$$

Moment Balance Equation:

$$-(F_{vib} \times \cos 65^\circ \times L/2 \times 22.22) + (mg \times 22.22) - (F_r \times \sin 5^\circ \times 22.22) + (F_s \times 45.63) - (R_2 \times 45.63) = 0$$

AISI 4030 Steel annealed
at 86.5 degree

[4130 Alloy Steel Sheet at
Rs 90/kg | Mumbai Central
| Mumbai | ID:
20319826862](#)

	$(-392.51 \times \cos 65^\circ \times 37.35 \times 22.22) + (9.81 \times 7.96 \times 22.22) - (16.27 \times 0.087 \times 22.22) + (30.2148 \times 45.63) - (R_2 \times 45.63) = 0$ Calculation steps: $15.0167 + 18.8435 - 0.3428 + 13.787 - R_2 \times (45.63)/100 = 0$ Final Results: Now solving force and moment balance equation we get the following results: $R_1 = 103.98 \text{ N} \quad R_2 = 103.98 \text{ N}$ <u>Bending Moment of separating plates:-</u> Total Mass=M= mass of plate + bolt + nuts = $m_1 = 5.21 + 2.25/2 + 0.5 = 6.835 \text{ kg}$ Force by each spring on plate= 103.98 N Fvib= 337.29 N r(1)= location of COM r(2)= distance of 1st spring from 2nd M1= Bending moment of the plate On doing analysis from the 2nd spring, the equation is as follows:- $mgr(1) - 103.98 \times r(2) - F_{vib} \times 0.2241 \times \sin(25) + M1 = 0$ $\Rightarrow 6.835 \times 9.81 \times 0.2241 - 103.98 \times 0.4463 - 337.29 \times 0.2241 \times 0.442 + M1 = 0$ $\Rightarrow M1 = 64.789 \text{ Nm (clockwise)}$ On doing analysis from the center of mass, we obtain that the bending moment is zero, showing that the mechanism is stable.	ASTM A36: Good Tensile strength Excellent weldability due to low carbon content Cheaper than SS MS is Preferred to SS and Iron as it has more yield strength and is more corrosion resistant respectively. ASTM A36 MS is less corrosion resistant than SS but can be galvanized. ASTM A36 MS is cheaper than SS304 ASTM A36 MS Beam price is approx 60-85 Rs /kg
--	--	---

Column Calculations:-

1.Buckling

Fixed end $k=0.5$; $E=200\text{GPa}$

$$I = (a_2^4 - a_1^4)/12$$

(a_2 =outer side length ; a_1 =inner side length)

$$I(\text{length})=0.3 \text{ m}$$

$$P_{cr}=(\pi^2 * E * I)/(kL)^2$$

As max load is 104 N(front column load)

By keeping FOS as 1.5

$$104 * 1.5 = (\pi^2 * E * I)/(kL)^2$$

And as $a_2=25.4 \text{ mm}$

Solving for a_1 ;

$$a_1=24.8 \text{ mm}$$

So wall thickness is 0.3 mm

As the standard industry thickness starts from 2mm. We take wall thickness as 2mm.

2.Compressive stress analysis

Beam dimension: (25.4mm x 25.4mm) square hollow beam (thickness 2mm)

$$\text{Beam area} = 187.2 \text{ mm}^2$$

Stress on square beam

$$= 5.556 \times 10^5 \text{ N/m}^2$$

Yield Strength of ASTM A36 MS square beam

$$= 2.5 \times 10^8 \text{ N/m}^2$$

As yield strength > Stress on Beam

so we can proceed

Long Term Wear:-

Archard's Wear Equation:

$$V = (K * F * s) / H$$

• Inputs and Assumptions:

- Bolt mass: 0.09 kg
- Plate angle: 5°
- Number of bolts: 25/min
- Sliding distance: 0.75 m/bolt
- Material: SS304 (Hardness = 200 MPa)

Calculation Steps:

- 1. Normal Force: $F = mg \cos(\alpha) = \mathbf{0.8826 \text{ N}}$
- 2. Total Sliding Distance: $s_{\text{total}} = 25 \times 0.75 = \mathbf{18.75 \text{ m/min}}$
- 3. Wear Volume: $V = (K * F * s_{\text{total}}) / H = \mathbf{1.65 \times 10^{-10} \text{ m}^3/\text{min}}$
- 4. Thickness Loss: $\Delta h = V / A$, $A = 0.075 \text{ m}^2 \rightarrow \Delta h = \mathbf{2.2 \times 10^{-9} \text{ m/min}}$

Results:

- Plate Thickness Loss (8 hrs/day, 250 days/year): $\Delta h_{\text{year}} = \mathbf{0.0264 \text{ mm/year}}$
- For 5 years: $\Delta h_{\text{5years}} = \mathbf{0.13 \text{ mm}}$

- Long-term wear is minimal; periodic inspection is recommended.

	<u>Spring Calculations:-</u> → $F_1 = F_2 = 104 \text{ N}$ Factor of safety = 1.5 The a priori decisions are: Music wire A228; ends squared and grounded; $F_{\max} = 104 \text{ N}$ design factor of safety = 1.5 Robust Linearity: $\xi = 0.15$ Starting with $d = 1.75 \text{ mm}$ and $y_{\max} = 2\text{mm}$ $m = 0.145$ relative cost = 2.6 $A = 2211 \text{ Mpa} \cdot (\text{mm})^m$ $E = 196,500 \text{ N/mm}^2$ $G = 81,000 \text{ N/mm}^2$ $S_{ut} = 2038.676 \text{ Mpa}$ $S_{sy} = 917.4 \text{ Mpa}$ $\alpha = 611.6 \text{ Mpa}$ $\beta = 99.447 \text{ Mpa}$ $C = 4.66$ $D = C \times d = 8.1554\text{mm}$ Bergstrasser factor $K_B = 1.319$ $\tau = 611.6\text{Mpa}$ $n_s = 1.5$ OD = 9.9 mm ID = 6.4 mm No. of active coils, $N_a = d^4 G / 8 D^3 k$ $N_a = 3.366$ coils	Assuming, 2 mm is to vibrate and move the bolts/grains. (Assuming $1.6 < d \leq 3 \text{ mm}$) Music wire, as-wound spring
Table 10-5		
Eq 10-14		
Table 10-6		
Eq 10-16		
Eq. 10-23		
Eq 10-1		
Eq 10-5		
Figure 10-3		
Eq 10-9		

Table 10-1	$N_t = N_a + 2 = 5.366$ coils $L_s = 9.3917$ mm $L_o = 11.6917$ mm Absolute stability condition $L_o < \pi D / \alpha \sqrt{(2(E-G))/(2G+E)} = L_{cr}$	End condition between parallel surfaces
Table 10-2	$\alpha = 0.5$	
Eq 10-13	$\Rightarrow L_{cr} = 42.897$ mm	
	We repeat this process with different values of d for the same music wire.	
	We then also consider an Oil Tempered wire and repeat the same calculations.	
	(relative cost factor = 1.3)	
	Considering the recommended design conditions:	
Eq 10-18	$4 \leq C \leq 12$	
Eq 10-19	$3 \leq N_a \leq 15$	
Eq 10-20	$\xi \geq 0.15$	
Eq 10-21	$n_s \geq 1.2$	
	d = 1.70 and 1.75 mm satisfy these conditions. Since, the cost for a 1.70 mm diameter would be lesser we can select a spring with d = 1.70 mm made of music wire.	
	If cost is an important aspect, we can go with a spring with d = 1.90 mm made of oil tempered wire, which would be the cheapest option satisfying our design conditions.	

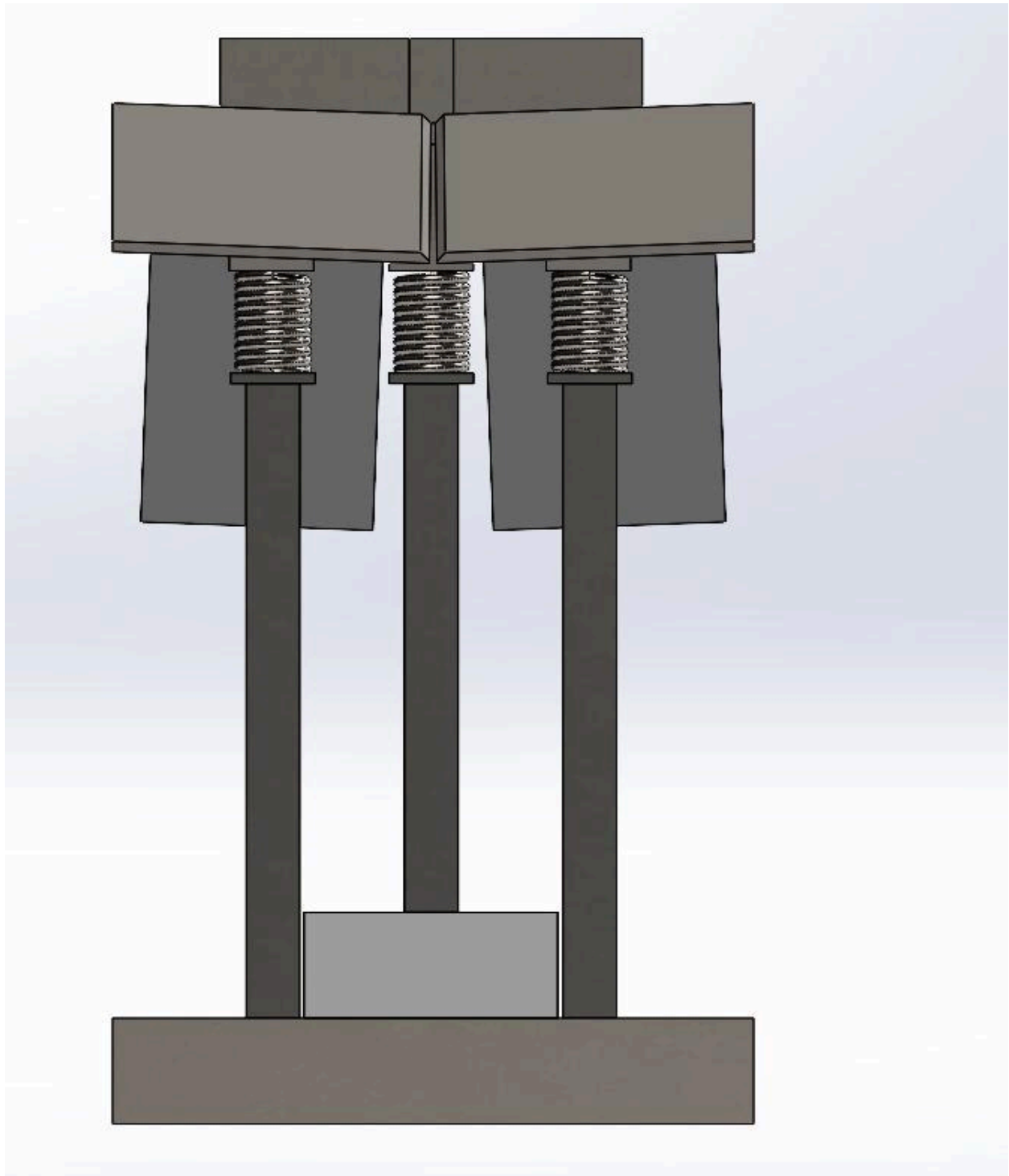
Music Wire:

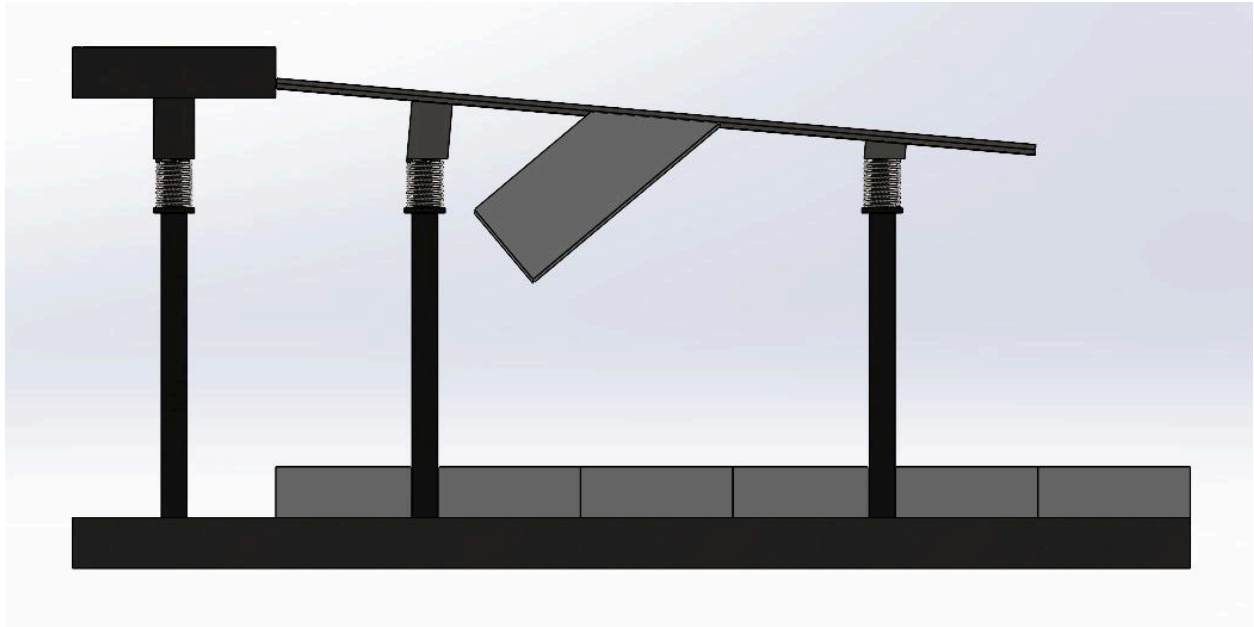
d (mm)	1.65	1.7	1.75	1.78	1.8
D	6.55	7.33	8.155	8.66	9.01
C	3.97	4.31	4.66	4.86	5
OD	8.2	9.03	9.9	10.44	10.81
N _a	5.122	4.11	3.366	3	2.78
L _s	11.75	10.4	9.39	8.9	8.61
L ₀	14.05	12.7	11.69	11.2	10.91
L _{cr}	34.48	38.58	42.89	45.59	47.43
n _s	1.5	1.5	1.5	1.5	1.5
fom	-815.488	-832.31	-859.89	-881.20	-897.326

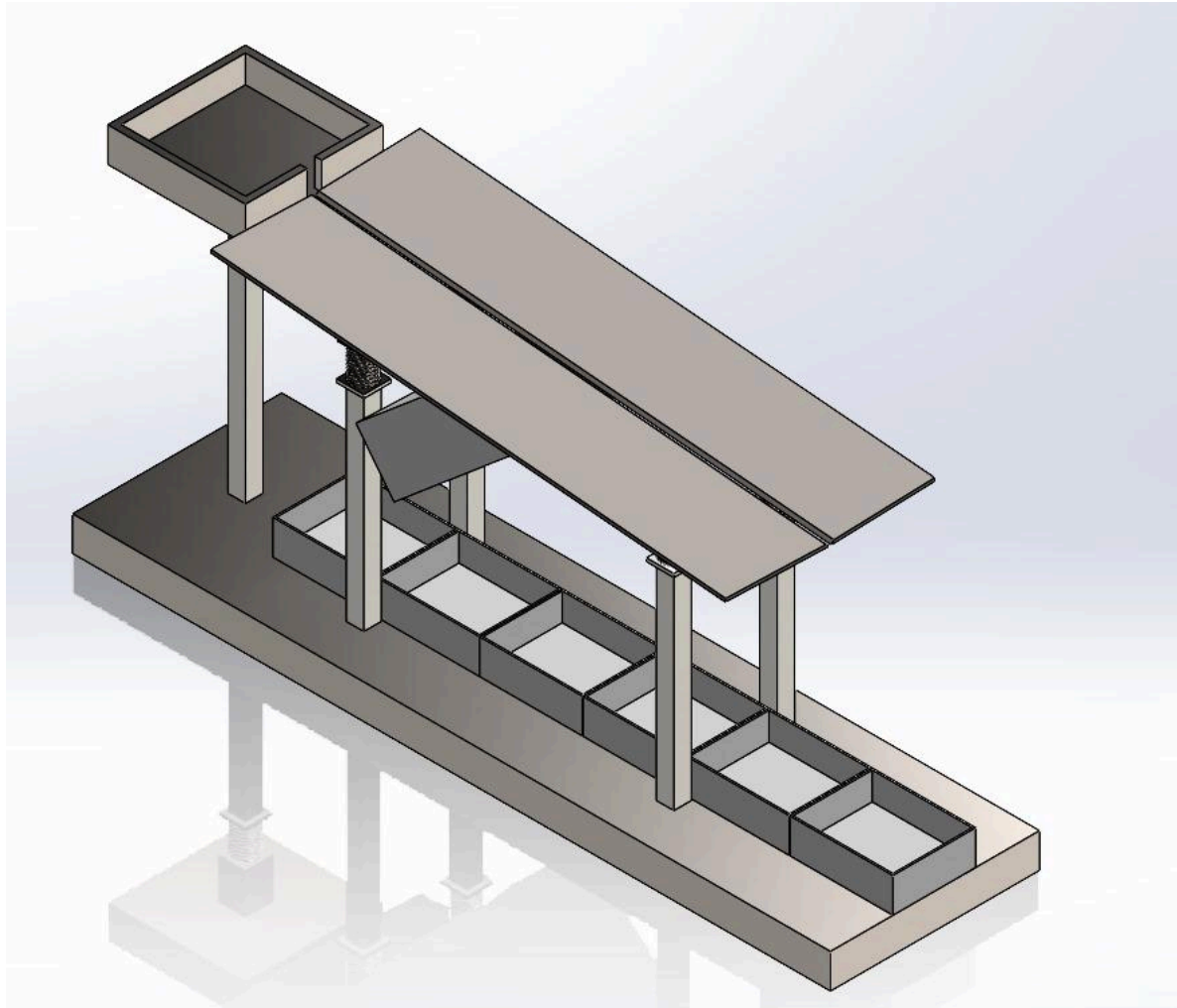
Oil Tempered:

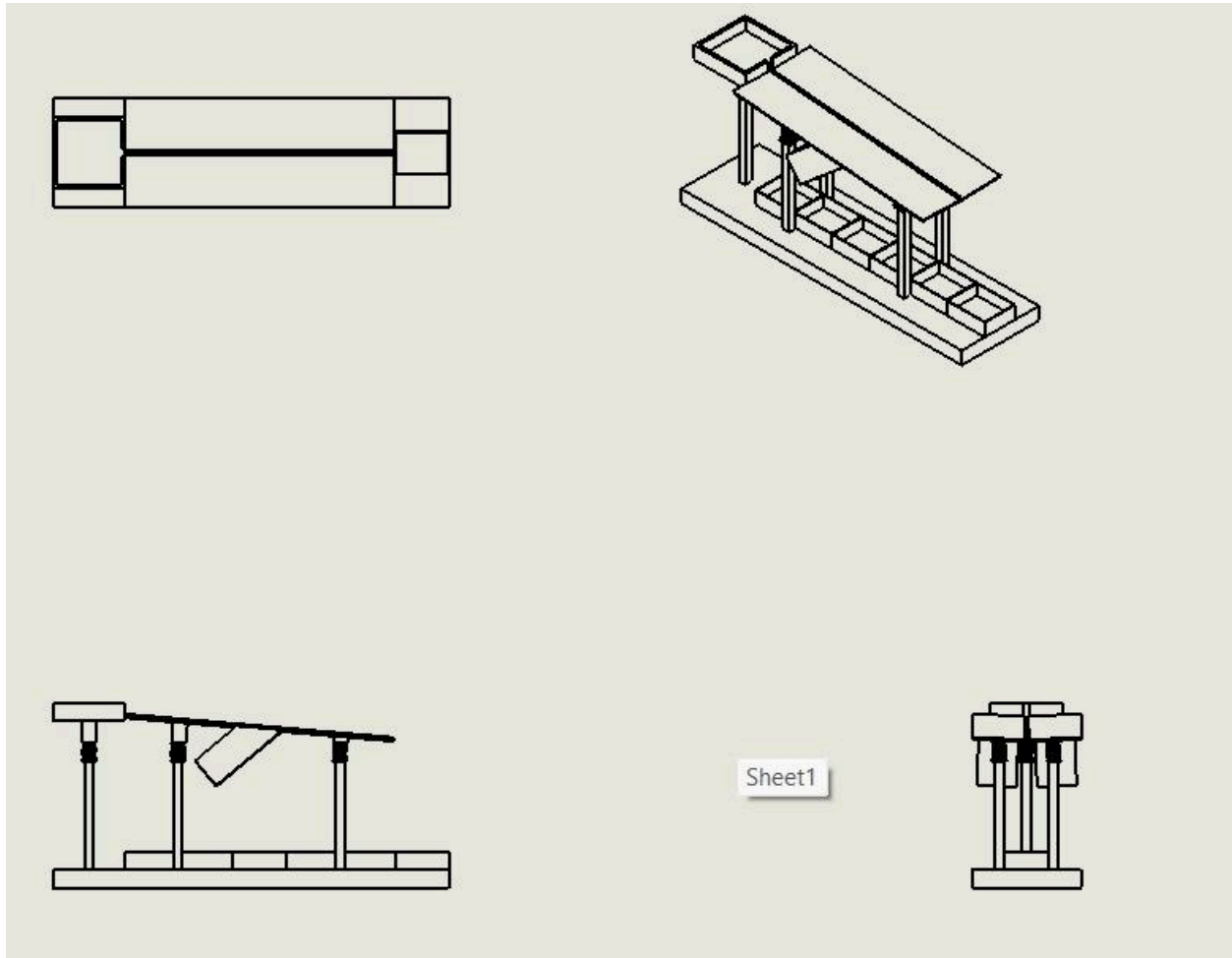
d (mm)	1.8	1.85	1.9	1.95	2.05
D	6.73	7.47	8.244	9.05	10.77
C	3.74	4	4.33	4.64	5.26
OD	8.53	9.322	10.144	11	12.825
N _a	6.55	5.35	4.433	3.717	2.69
L _s	15.4	13.6	12.223	11.14	9.61
L ₀	17.7	15.9	14.525	13.44	11.91
L _{cr}	35.41	39.3	43.364	47.607	56.678
n _s	1.5	1.5	1.5	1.5	1.5
fom	-598.71	-603.08	-614.17	-631.17	-681.36

CAD DRAWING OF CONCEPT:









Link to CAD Model: [CAD MODEL](#)

ANIMATION: [DME Project Animation](#)

Conclusion: The sorting machine is able to sort nuts and bolts according to their size as they move down the inclined plate with the help of gravity and the vibration force.

